

AIRPORT FLIGHT LOOKUP SYSTEM USING BINARY SEARCH

1. Objective

To analyze how Binary Search improves the efficiency of searching flight records in an airport management system and implement the algorithm using Python.

2. Problem Statement

Airport management systems store thousands of flight records. Efficient retrieval of flight information is necessary to provide quick assistance to passengers.

3. Introduction

Binary Search is an efficient searching algorithm that works on sorted data by repeatedly dividing the search space into two halves.

4. Requirements

- Flight records must be sorted.
- Fast response time is required.
- Suitable for large datasets.
- Supports random access.

5. Algorithm

Step 1: Initialize low and high.
Step 2: Find the middle element.
Step 3: Compare the target with the middle element.
Step 4: Search the left or right half accordingly.
Step 5: Repeat until found or search ends.

6. Python Program

Binary Search is implemented using a loop that continuously reduces the search range until the target flight number is found.

7. Sample Input

AI525

8. Sample Output

Flight AI525 found at index 4.

9. Advantages

- Faster than Linear Search.
- Reduces passenger waiting time.
- Suitable for large databases.
- Improves airport service efficiency.

10. Conclusion

Binary Search reduces search complexity from $O(n)$ to $O(\log n)$, making it ideal for airport flight lookup systems.

Time Complexity Analysis

Case	Time Complexity
Best Case	$O(1)$
Average Case	$O(\log n)$
Worst Case	$O(\log n)$